PROJECT PROPOSAL & ABSTRACT

Smart Wireless Fast-Charging and Regenerative Braking Integration for Electric Vehicles

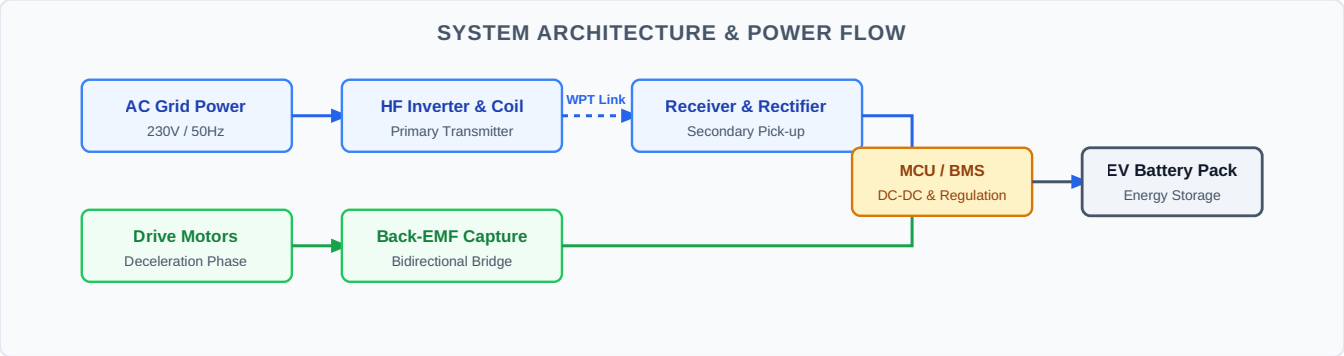
Domain: Electric Mobility • Embedded Systems • Energy Harvesting

METHODOLOGY

This project develops an integrated energy-harvesting and wireless power transfer framework for Electric Vehicles (EVs) aimed at extending operational range and eliminating physical charging bottlenecks:

- **Resonant Inductive Wireless Power Transfer:** High-frequency magnetic resonance transfers electrical energy wirelessly from a ground transmitter pad to an underbody vehicle receiver coil, followed by active synchronous rectification and buck-boost regulation for rapid battery replenishment.
- **Microcontroller-Governed Regenerative Braking:** During braking and deceleration cycles, counter-electromotive force (back-EMF) from the drive motors is harvested via bidirectional H-bridge switching circuitry, conditioned, and routed back to recharge the energy storage system.
- **Intelligent Energy Management (BMS):** An onboard microcontroller continuously supervises battery state of charge (SoC), thermal profiles, and coil proximity, orchestrating seamless power-source transitions between kinetic energy recovery and wireless charging.

PROPOSED BLOCK DIAGRAM



PROPOSED OUTCOME

- **Extended Operational Range:** Recaptures kinetic deceleration energy to boost overall vehicular energy utilization efficiency by 15–20%.
- **Autonomous Hands-Free Fast Charging:** Eliminates mechanical wear-and-tear and manual connector risks via high-efficiency resonant inductive charging.
- **Dynamic State Supervision:** Embedded sensors provide autonomous overcharge and overcurrent safeguarding during high-rate power transfer.
- **Prototype Demonstration:** Implements a hardware test vehicle validating steady power transition between regenerative recovery and wireless docking.